

Tap. Validate. Welcome.

A human-first blueprint for NFC check-in and freebie collection
- operated by Axon, resilient by design.

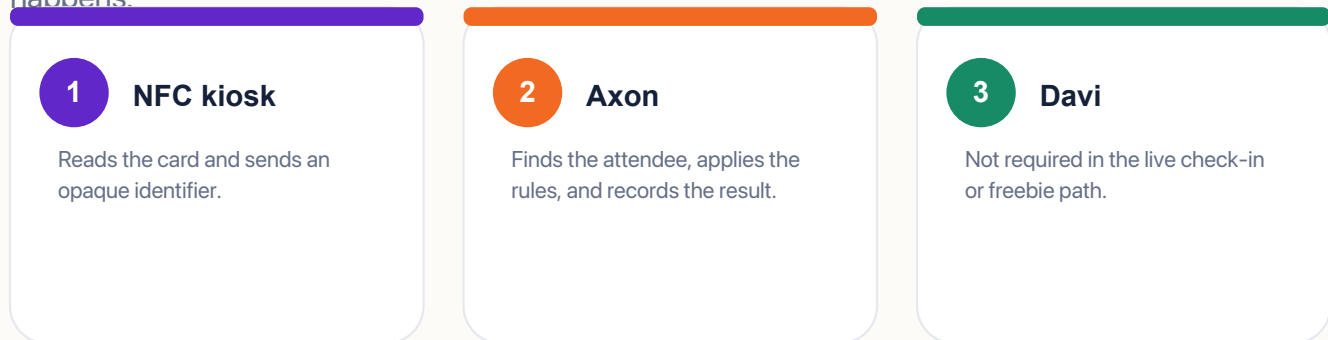
THE DECISION

Davi contributes the NFC kiosk repository. Axon owns the live transaction.

No Davi server is required to approve check-in or release a freebie.

The architecture becomes simpler - and safer.

NFC is the way an attendee presents identity. Axon remains the system that decides and records what happens.



ONE-SENTENCE OPERATING MODEL

The attendee taps the card; Axon recognizes the person and records the action.

If NFC is unavailable, the same attendee can still use the existing QR ticket or staff-assisted search.

WHY THIS IS THE RIGHT JULY SCOPE



Fewer dependencies

Admission does not wait for another platform.



One source of truth

Axon owns attendance and collection records.



Graceful fallback

QR and manual lookup remain available.

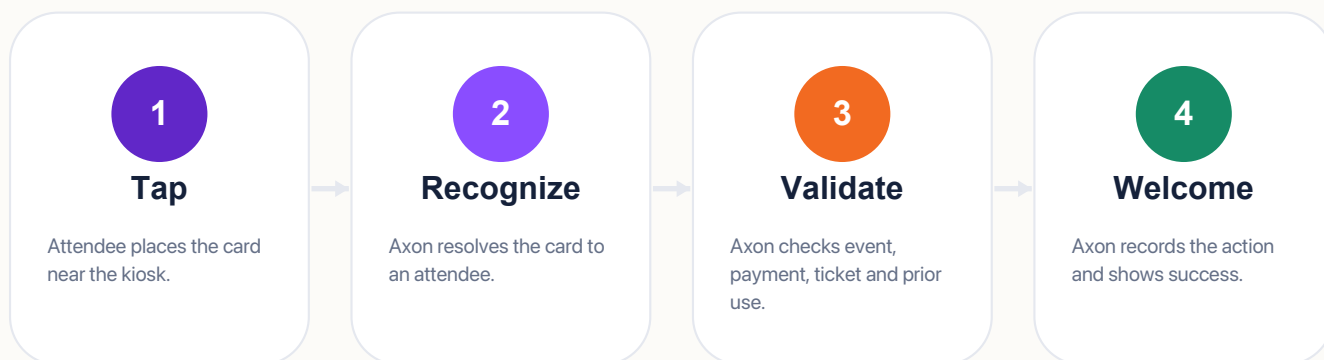


Clear accountability

Event staff know exactly which system decides.

A fast front-door experience with three safety nets.

The design optimizes for the happy path without trapping staff when hardware, connectivity, or a card fails.



WHEN A TAP DOES NOT WORK

01

Scan the Axon QR

02

Search name, email or reference

03

Use a controlled manual exception

WHAT STAFF SHOULD SEE

SUCCESS

Juan Dela Cruz checked in

DUPLICATE

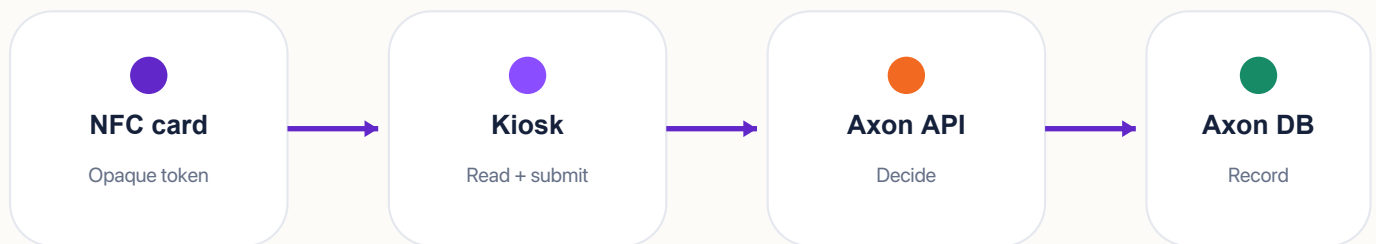
Already checked in at 8:15 AM

ACTION

Card not linked - search attendee

Davi is outside the live decision loop.

This is the most important boundary in the design.



LIVE TRANSACTION BOUNDARY

Card tap -> Axon validation -> Axon record

There is no synchronous Davi approval, webhook, identity lookup, or redemption call here.

DAVI CONTRIBUTION

NFC kiosk repository

Code handover, card-format documentation and implementation guidance.

NOT IN SCOPE FOR THE TAP

Davi production availability

A Davi outage must not prevent entry or freebie collection.

A tap is a request - not automatic permission.

Axon must still apply every admission rule before recording attendance.

- 1 Card active?**
Reject revoked or replaced cards.
- 2 Correct event?**
A card link does not grant entry everywhere.
- 3 Payment confirmed?**
Registration must be verified or order paid.
- 4 Ticket valid?**
Reject cancelled, refunded or invalid tickets.
- 5 First check-in?**
Prevent repeated or simultaneous admission.

ATOMIC CHECK-IN

Only one winner.

If two kiosks submit the same attendee at nearly the same moment, the database must accept one check-in and return the first result to the other.

REQUIRED UNIQUENESS

One accepted check-in per attendee + event.

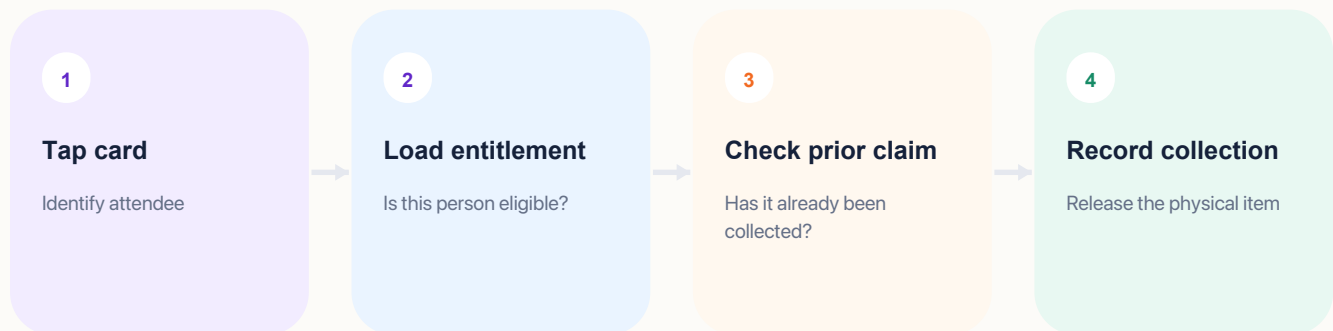
EXISTING AXON ADVANTAGE

The QR scanner already validates event, registration status and duplicate attendance.

The NFC endpoint should reuse the same business rules and produce the same audit result - only the input method changes.

The same tap, a different rulebook.

Check-in proves arrival. Freebie collection proves an entitlement was consumed. They must be stored as separate events.



NON-NEGOTIABLE CONTROL

One accepted claim per attendee + event + freebie.

Repeated taps, retries, double-clicks and simultaneous stations must return the original result - never issue a second item.

WHAT THE FREEBIE RECORD SHOULD ANSWER

- Who received it?
- Which station and staff member?
- Was an override used - and why?
- What item was released?
- When was it collected?
- Was the action online or provisional?

NFC hardware is common. Access is not universal.

The phone may contain an NFC radio while the chosen browser or application still cannot use it in the required way.

ANDROID KIOSK

Strong web option

-  Chrome for Android supports Web NFC.
-  NDEF-compatible cards only.
-  Requires HTTPS and permission.
-  Page visible; device unlocked.
-  Test exact phone, OS and card.

IPHONE KIOSK

Native path recommended

-  iPhones contain NFC hardware.
-  Do not assume Safari Web NFC.
-  Use a native Core NFC app.
-  Or use a dedicated NFC reader.
-  Verify entitlement and card protocol.

JULY RECOMMENDATION

Use a fixed, tested device fleet.

If the repository is browser-based Web NFC, prefer managed Android phones running Chrome. Keep QR as the universal fallback.

Needs verification: kiosk framework, iOS support, NFC protocol, NDEF compatibility, card write protection and use of card serial numbers.

Offline mode is a controlled exception, not magic.

A local cache can keep one station moving. It cannot perfectly coordinate multiple disconnected entrances by itself.

ONLINE - PREFERRED

Server decides immediately.

- 1 Read card
- 2 Call Axon
- 3 Apply current rules
- 4 Write final result
- 5 Return success

OFFLINE - PROVISIONAL

Kiosk records intent locally.

- 1 Read card
- 2 Check encrypted cache
- 3 Prevent same-device repeat
- 4 Queue signed event
- 5 Sync and reconcile

HARD TRUTH

Two disconnected kiosks can both accept the same card.

For July, designate one offline entrance or provide a local venue network.

FREEBIES NEED STRICTER FALLBACK

One station

Only one offline station per freebie type.

Visible tally

Staff tracks physical stock and provisional claims.

Supervisor

Offline exceptions require named approval.

The card points to identity. It should not carry identity.

Treat the NFC card like a claim ticket: useful only when Axon validates it.

SAFE ON THE CARD

Opaque random token

or a signed Axon card token

No name. No email. No payment state. No entitlement list.

NEVER ON THE CARD

-  **Personal details**
-  **Payment information**
-  **QR signing secret**
-  **Staff credentials**
-  **Full attendee record**

KIOSK LEAST PRIVILEGE

A kiosk account should only check in or redeem.

Station identity

Every device is registered.

Short-lived access

Credentials expire quickly.

Event scope

Only the assigned event.

Action scope

Check-in or freebie only.

Remote revoke

Disable a lost device.

Encrypted cache

Protect offline attendee data.

Small APIs. Strong guarantees.

The public experience stays simple because the server contract handles identity, concurrency, audit and retries.

NFC CHECK-IN REQUEST

```
POST /event-operations/check-ins/nfc
{
  "eventId": "evt_...",
  "cardToken": "opaque-token",
  "stationId": "entrance-a-01",
  "operationId": "unique-id",
  "capturedAt": "ISO-8601"
}
```

IDEMPOTENCY

Retrying the same operation returns the same result.

It must not create another check-in or claim.

AUDIT

Every tap produces a traceable outcome.

Station, staff, time, mode, decision and override reason.

MINIMUM NEW RECORDS

`nfc_cards`

Card token and lifecycle

`attendee_card_assignments`

Card-to-person link

`check_in_events`

Immutable attempts/results

`event_stations`

Kiosk identity and scope

`attendee_entitlements`

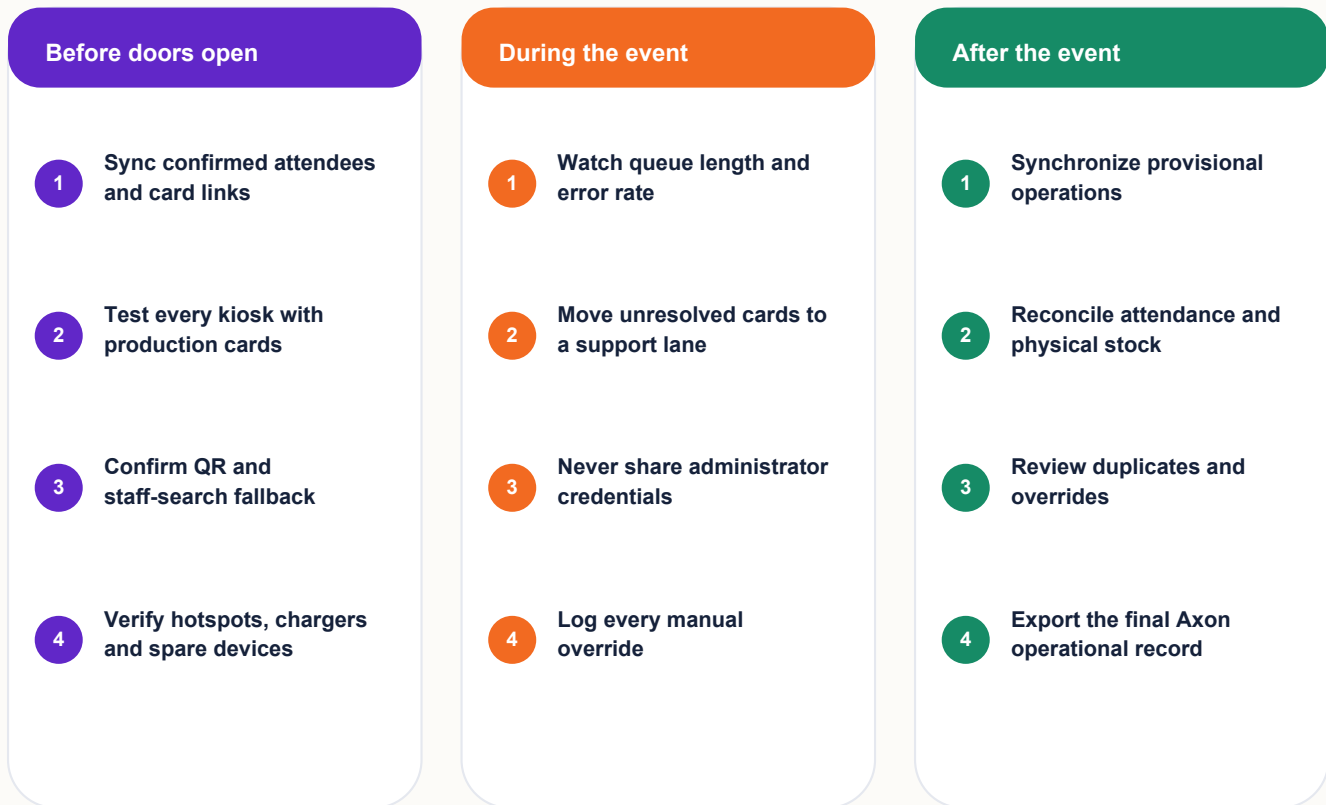
Who can claim what

`freebie_redemptions`

Collection ledger

Design for the queue, not the demo.

A production-ready kiosk is equal parts software, hardware, staffing and fallback discipline.



INCIDENT COMMAND

One person owns the operational decision.

Axon handles admission and freebies. Davi supports repository/card-format questions. Staff escalate through one named event lead.

NFC

QR FALLBACK

MANUAL SEARCH

OFFLINE EXCEPTION

RECONCILE

GO WITH CONDITIONS

The revised design is safer, but the NFC implementation must be proven on the exact repository, cards and kiosk devices.

MINIMUM SAFE JULY SCOPE

- | | |
|---|--|
|  Review the contributed kiosk repository |  Confirm card protocol and NDEF compatibility |
|  Direct kiosk-to-Axon integration |  Atomic, idempotent check-in |
|  QR and manual fallback |  One controlled freebie pilot |
|  Device-specific rehearsal |  Named offline procedure |

DEFER

- Replacing QR entirely
- Mixed untested device fleet
- Multi-station offline freebies
- Virtual cards on attendee phones

REPOSITORY QUESTIONS

- ? Web, Android, iOS or cross-platform?
- ? Which NFC protocols are read?
- ? Does it use UID or NDEF payload?
- ? How does offline storage work?

DECISION RULE

If NFC fails, the event must still run.

That is why NFC enhances Axon check-in; it does not replace the proven QR and staff-assisted paths for July.

FINAL RECOMMENDATION

Keep the moment simple.

Tap the card. Let Axon decide. Keep QR ready.

Davi provides the NFC kiosk foundation.

Axon owns identity resolution, eligibility, check-in, freebie collection, offline recovery, audit and reporting.

That boundary minimizes event-day failure, protects attendee data and keeps accountability clear.

PRIMARY REFERENCES

Chrome Web NFC: <https://developer.chrome.com/docs/capabilities/nfc>

Apple Core NFC: <https://developer.apple.com/documentation/corenfc>

MDN NDEFReader: <https://developer.mozilla.org/en-US/docs/Web/API/NDEFReader>

Axon evidence: NestJS services, Prisma schema, QR utilities and admin check-in UI reviewed 20 June 2026.